Fiscal Year:	FY 2024	**Task Last Updated:**	FY 01/31/2024
PI Name:	Patla, Bijunath Ph.D.		
Project Title:	NIST/ACES Collaboration: ACES Ground Station and Data Analysis Center in Boulder, Colorado		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	brp1@nist.gov	**Fax:**	FY
PI Organization Type:	GOVERNMENT	**Phone:**	303-497-4082
Organization Name:	National Institute of Standards and Technology		
PI Address 1:	Time & Frequency Division, MS 847		
PI Address 2:	325 Broadway Street		
PI Web Page:			
City:	Boulder	**State:**	CO
Zip Code:	80305	**Congressional District:**	2
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
Start Date:	04/01/2021	**End Date:**	09/29/2028
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Callas, John	**Contact Phone:**	
Contact Email:	john.l.callas@jpl.nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/29/2028 per D. Griffin/NASA-HQ (Ed., 1/20/25).		
Key Personnel Changes/Previous PI:	New Principal Investigator as of April 1, 2021 is Dr. Bijunath R. Patla; previous PI was Dr. Chris Oates.		
COI Name (Institution):			
Grant/Contract No.:	NNH12AT81I		
Performance Goal No.:			
Performance Goal Text:			
	NOTE: Continuation of "NIST/ACES Collaboration: ACES Ground Station and Data Analysis Center in Boulder, Colorado (PI=Oates)" with new PI, Dr. Bijunath R. Patla. To support primary Atomic Clock Ensemble in Space (ACES) mission objectives, the National Institute of Standards and Technology (NIST) proposes to: 1. Develop a Microwave Link Ground Terminal (MWL GT) at NIST in Boulder, CO to perform 10^-17 uncertainty time and frequency comparisons between ACES clocks and world-leading optical and atomic clocks at NIST Boulder and the nearby JILA (formerly known as the Joint Institute for Laboratory Astrophysics); and to support general ACES science goals.		

Task Description:

2. Participate as fully as possible in coordinated European Space Agency (ESA) data analysis with formal ESA Data Analysis Centers and other ACES partners throughout the world. NIST will collaborate closely with ESA partners to define how NIST can best contribute to overall ACES mission goals within the constraints of NIST resources.

As part of participation in the time and frequency transfer and data analysis, NIST will participate in tests of fundamental physics based on ACES data to include at least the following:

• Global time and frequency transfer with uncertainty as low as 10^-17.

• Direct comparison of PHARAO space clock to NIST time and frequency standards at uncertainty of 10^-17 or better.

• Test of gravitation redshift at 2 x 10^-6 uncertainty or better.

• Search for variation in fundamental constants, such as fine structure constant at level of 10^-18 per year or better.

All these goals will be facilitated by the strong time and frequency infrastructure at NIST, including (a partial list):

• World-class cesium microwave primary frequency standards, including NIST-F1 with fractional frequency uncertainty of 3 x 10^-16, and NIST-F2 which will be operational in 2013 with an eventual uncertainty of 1 x 10^-16 or better.

• Two aluminum ion “logic” optical clocks with current fractional frequency uncertainty of approximately 9 x 10^-19.

• Two ytterbium lattice optical clocks with uncertainties better than 2 x 10^-18.

• Strontium lattice optical clocks with uncertainties of 2 x 10^-18 (at JILA). A dedicated fiber optic system connects NIST and JILA to enable the JILA clocks to directly participate in NIST time and frequency transfer and intercomparisons.

• A high performance calcium optical frequency standard with exquisite short-term stability to serve as a “fly wheel” for clock comparisons with demonstrated instability below 10^-16.

• A world-class time scale comprising eight hydrogen masers and a collection of Cs clocks.

NIST scientists will participate in ESA/NASA planning meetings about the ACES program in general, and will specifically consult with ESA to determine how NIST can most fully participate in MLW GT data analysis and applications.

In addition to the NIST resources already proposed for this program (NIST and JILA optical and microwave frequency standards, NIST time scale, NIST time and frequency transfer capabilities, etc.) NIST will directly contribute substantial funds and expert NIST staff time significantly exceeding the direct NASA investment.

We propose the following revised program plan outline:

Year 1 (2012): Begin design and construction of platform and other infrastructure for MWL hardware. $150k for partial payment of platform and related infrastructure design and construction.

Year 2 (2013): Complete construction of platform and other infrastructure for MWL system. $150k for partial payment of platform and related infrastructure.

Year 3 (2014): Receive MWL hardware. Install, test, and integrate MWL hardware into NIST time and frequency transfer system. Prepare and test MWL system for ACES time and frequency transfer. New NIST staff to prepare for participation in MWL data analysis. $150k for partial support of staff time.

Year 4 (2015): ACES launch. Participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. $150k for partial support of staff time.

Year 5 (2016): On-going ACES experiments. Participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. $150k for partial support of staff time.

Gap Year (2017): Perform relativistic calculations to prevent unanticipated shifts from contaminating ACES data. Install updated MWL hardware into new NIST time and frequency transfer system. Prepare updated data analysis system in anticipation of ACES launch. $150k for partial support of staff time.

Gap 6 Mo. (2018): Develop and install precision time links between the MWL hardware site and NIST clocks. $80k for partial support of staff time.

Gap 12 Mo. (2018-2019): Assuming successful launch in 2020, will be prepared to participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. Develop a frequency-based time scale for ACES frequency comparisons. $80k for partial support of staff time.

Gap 12 Mo. (2019-2020): Assuming successful launch in 2021, will be prepared to participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. Evaluate and improve the performance of the new frequency-based time scale optimized for ACES frequency comparisons. $80k for partial support of staff time.

Gap 12 Mo. (2020-2021): Evaluate and improve the performance of the new frequency-based time scale. Upgrade hardware infrastructure to add resiliency and robustness to the frequency-based time scale. $80k for partial support of staff time.

Gap 13 Mo. (2021-2022):Develop and integrate methods for precise orbit determination. Accurately estimate higher-order relativistic effects,estimate time and frequency transfer noise for clocks in orbit. $90k for partial support of staff time.

Gap 12 Mo (2022-2023) Assess relevant issues such as the adequacy of knowledge of the geopotential (including dynamical effects) for supporting world-wide time comparisons with advanced clocks. $80k for partial support of staff time.

Prepare for the public release of data products in collaboration with ACES/ESA and extend investigations to benefit future missions with more precise clocks for testing fundamental physics. $93k for partial support of staff time.

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The major benefits and impacts of this research program can be grouped into three main areas: 1. Substantial improvements in international coordination of extremely precise and accurate time and frequency information. Precision timekeeping and synchronization are crucial for such ubiquitous technologies as precision positioning and navigation, telecommunications, remote sensing, surveillance, and many other applications. The current technical limitations on international time and frequency transfer impose fundamental limits on these technologies. This project seeks to improve international time and frequency transfer by at least a factor of 10, with the potential for significantly greater improvements. Such improvements would not only directly enhance the current technologies depending on precision timing and synchronization, but would also open up new applications directly related to ultraprecise timing such as gravimetry, geodesy, magnetometry, and other measurements based on tiny perturbations in precision frequency measurements. 2. This project will serve as a pathfinder demonstration of the ability to perform cold atom experiments in space. Cold atoms and molecules represent a major, growing research opportunity for a wide variety of technologies, from ultraprecision measurements of all kinds (time, gravity, magnetic fields, electrical quantities, temperature, etc.) to the potential for development of exotic new materials. Having mastered cold atom and molecule experiments in the Earth laboratory, being able to perform such experiments in space opens the benefits of precision measurements in the space environment as well as microgravity experiments not possible in the Earth-based laboratory. 3. Fundamental physics experiments. The operation of a cold atom clock in space provides the opportunity to significantly improve measurements on the limits of special and general relativity and related phenomena, such as search for possible Lorentz transformation variations and search for possible time variations in fundamental constants, at levels 10 to 100 times more precise than previous experiments. Most major physics discoveries in recent decades have come from tiny variations in expected behavior, and these experiments would have profound implications on fundamental physics and cosmology.
Task Progress:	1. Theoretical framework for estimating higher order relativistic effects associated with frequency transfer, and understanding time-transfer between Earth and orbiting clocks. We continue with our analysis for comparing clocks in space and on the ground by integrating the methods and formalism for translating coordinate frequencies to proper times of orbiting clocks for testing fundamental physics: the principles of Local Position Invariance (LPI), Local Lorentz Invariance (LLI), and Universality of Gravitational Redshift (UGR). Estimating frequency transfer from a transmitter on a rotating frame to a receiver on a freely falling reference frame involves calculations; and developing simulations for the particular case of the International Space Station (ISS) as a freely falling frame, and possible other clocks in orbits and their ability to test fundamental physics, are progressing. Using the frequency scale algorithm, we minimized uncertainties and systematics on the ground clocks for comparing the fractional frequency of orbiting clocks. We also developed numerical routines for the orbit determination of space clocks to reduce uncertainties in space vehicles' position and velocity coordinates. A test suite for validating time and frequency transfer using at least three reference frames to translate observables from fixed coordinates to rotating coordinates was developed and tested. In part, we use past ISS orbits to simulate frequency transfer noise to constrain the above-mentioned fundamental physics variables. Anticipating a future of clocks in space, we are continuing with our efforts for estimating relativistic corrections for clocks in orbit and extending to L1, L2, and L4 Lagrange points. We consider various coordinate systems such as the rotating center of mass (Earth-Moon) coordinates, the rotating frame with the Earth at the origin, and the center of mass inertial coordinate system. 2. Design and development of continuously running frequency standards. We upgraded clock data monitoring and processing software to include National Institute of Standards and Technology (NIST) fountains and optical clocks with hydrogen masers (H-masers). Continuous monitoring and analyses seamlessly integrate the fountain data for actively steering the NIST timescale, and for providing periodic clock file updates to the Atomic Clock Ensemble in Space (ACES) data analysis center. After further upgrades, both NIST fountains, F3 and F4, are now in operation with evaluated uncertainties hovering around low 3.0E-15 in fractional frequency averaged over a day. We plan to further improve the short-term stability by integrating a single ion Sr+ clock for frequency corrections. This clock would primarily rely on two stabilized lasers -- one for cooling (422 nm) and another for clock transition (674 nm). The performance of UTC(NIST) -- real-time approximation of coordinated universal time (UTC) -- has significantly improved over the past four years as a result of frequently informed corrections in time and frequency, and other infrastructure upgrades. As a result, UTC(NIST) has stayed within +/- 2ns of UTC.
Bibliography Type:	Description: (Last Updated: 02/03/2026)
Abstracts for Journals and Proceedings	Hoth GW, Radnaev A, Mitchell SP, Sherman JA, Gergino V. "NIST-F3, a cesium fountain frequency reference." PTTI 2023 (Precise Time and Time Interval Systems and Applications Meeting), Long Beach, CA, January 23-26, 2023. Abstracts. PTTI 2023 (Precise Time and Time Interval Systems and Applications Meeting), Long Beach, CA, January 23-26, 2023. , Jan-2023
Articles in Peer-reviewed Journals	Parker TE, Brown RC, Sherman JA. "Statistics for quantifying aging in time transfer system delays." Metrologia. 2023 Nov 10;60(6):065011. https://doi.org/10.1088/1681-7575/ad088b , Nov-2023

Articles in Peer-reviewed Journals	Nardelli NV, Leopardi H, Schibli TR, Fortier TM. "Optical and microwave metrology at the 1E-118 level with an Er/Yb:glass frequency comb." Laser Photonics Rev. 2023 Apr;17(4):2200650. https://doi.org/10.1002/lpor.202200650 , Apr-2023
Articles in Peer-reviewed Journals	Collopy AL, Schmidt J, Leibfried D, Leibrandt DR, Chou CW. "Effects of an oscillating electric field on and dipole moment measurement of a single molecular ion." Phys. Rev. Lett. 2023 May 31;130(22):223201. https://doi.org/10.1103/PhysRevLett.130.223201 , May-2023
Articles in Peer-reviewed Journals	Ashby N, Patla B. "Simulations of the Hadamard variance: probability distributions and confidence intervals." IEEE Trans Ultrason Ferroelectr Freq Control. 2015 Dec 9;63(4):636-45. http://dx.doi.org/10.1109/TUFFC.2015.2507441 , Dec-2015
Articles in Peer-reviewed Journals	Ashby N, Parker TE, Patla BR. "A null test of general relativity based on a long-term comparison of atomic transition frequencies." Nature Physics. 2018 Aug;14(8):822-6. https://doi.org/10.1038/s41567-018-0156-2 , Aug-2018